
MODEL SELECTION AND VALIDATION REPORT (fishing_teve model)

OBSERVATIONS:1739

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (fishing_teve model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: gaussian(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	-0.01	0.08
log_sum_fishing	-0.08	0.03
sbt_mean	0.17	0.06
sbt_sd	0.01	0.04
sst_sd	0.02	0.03
sic_mean	-0.09	0.05
log_chla_mean	-0.04	0.03
depth	0.19	0.04

Dispersion parameter: 0.84

Spatiotemporal AR1 correlation (rho): 0.86

Matérn range: 167.25

Spatiotemporal marginal AR1 SD: 0.50

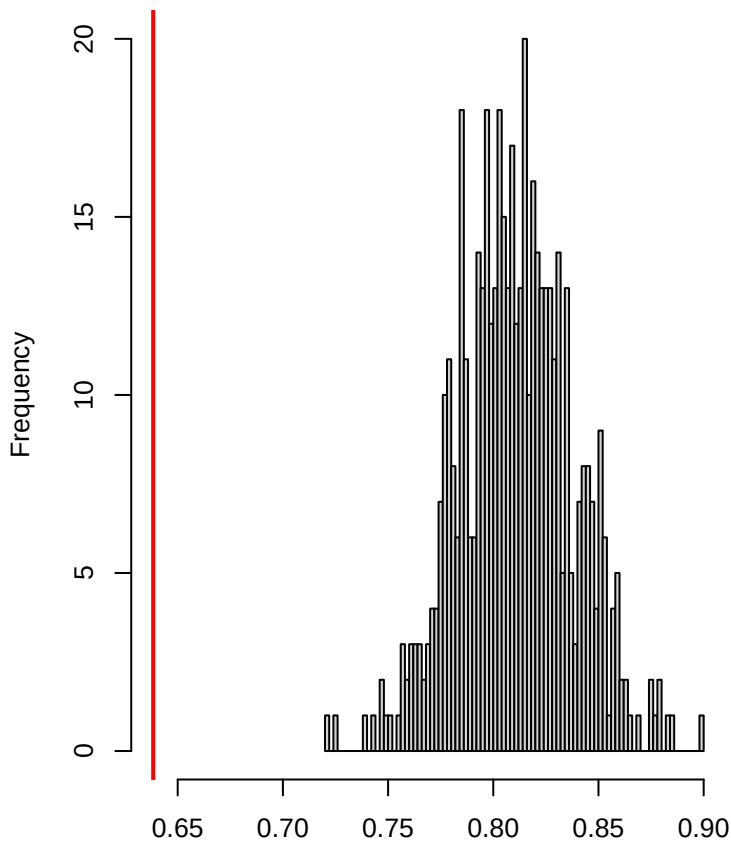
REML criterion at convergence: 2332.795

See ?tidy.sdmTMB to extract these values as a data frame.

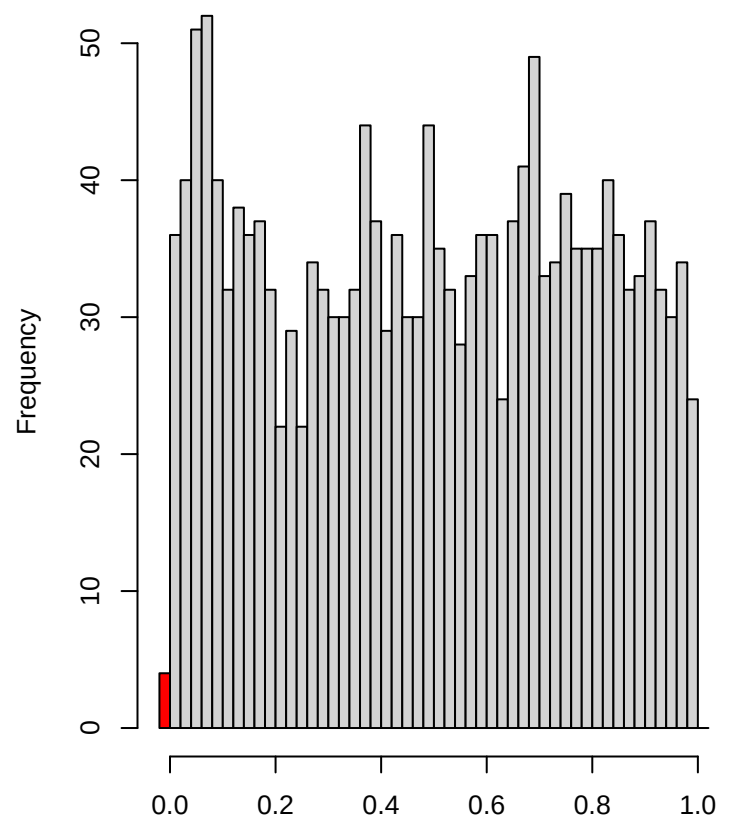
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	12	4689.590	2.223232
spatiotemporal_random_fields_ar1_with_anisotropy	14	4691.813	43.818562
spatial_random_fields	11	4735.632	2.459544
spatial_random_fields_with_anisotropy	13	4738.091	49.546601
spatiotemporal_random_fields_iid	11	4787.638	1.383969
spatiotemporal_random_fields_iid_with_anisotropy	13	4789.022	36.928557
error_only	9	4825.950	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



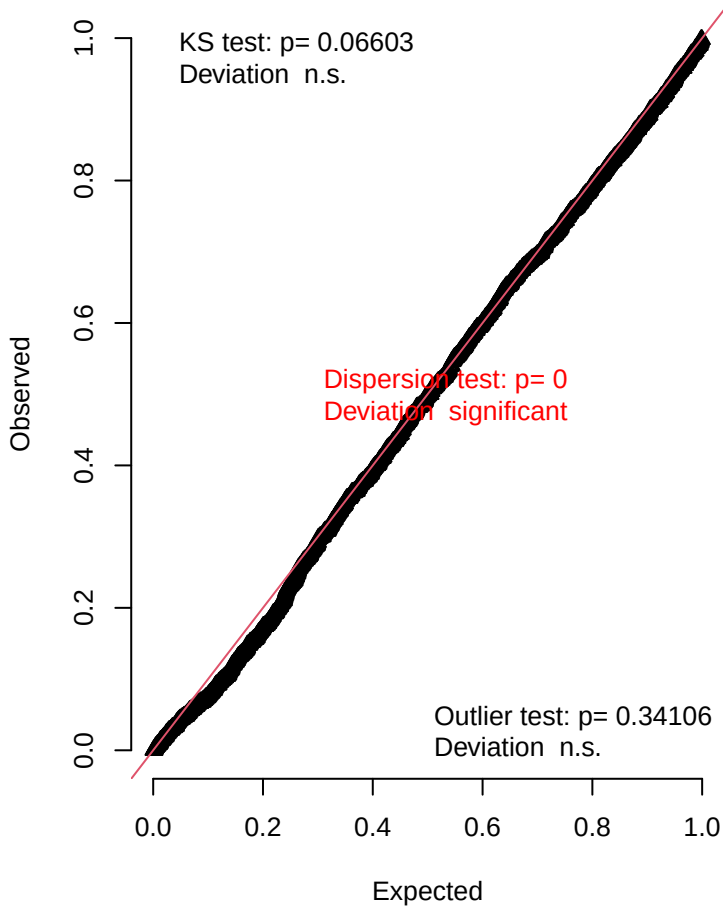
Outlier test n.s.



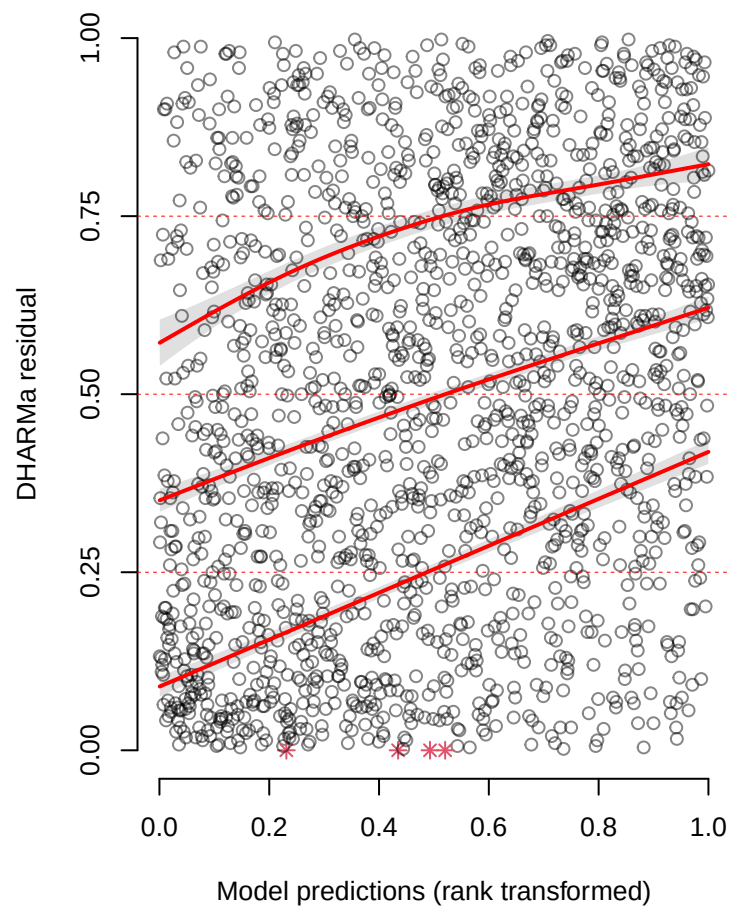
Simulated values, red line = fitted model. p-value (two.sided)

Residuals (outliers are marked red)

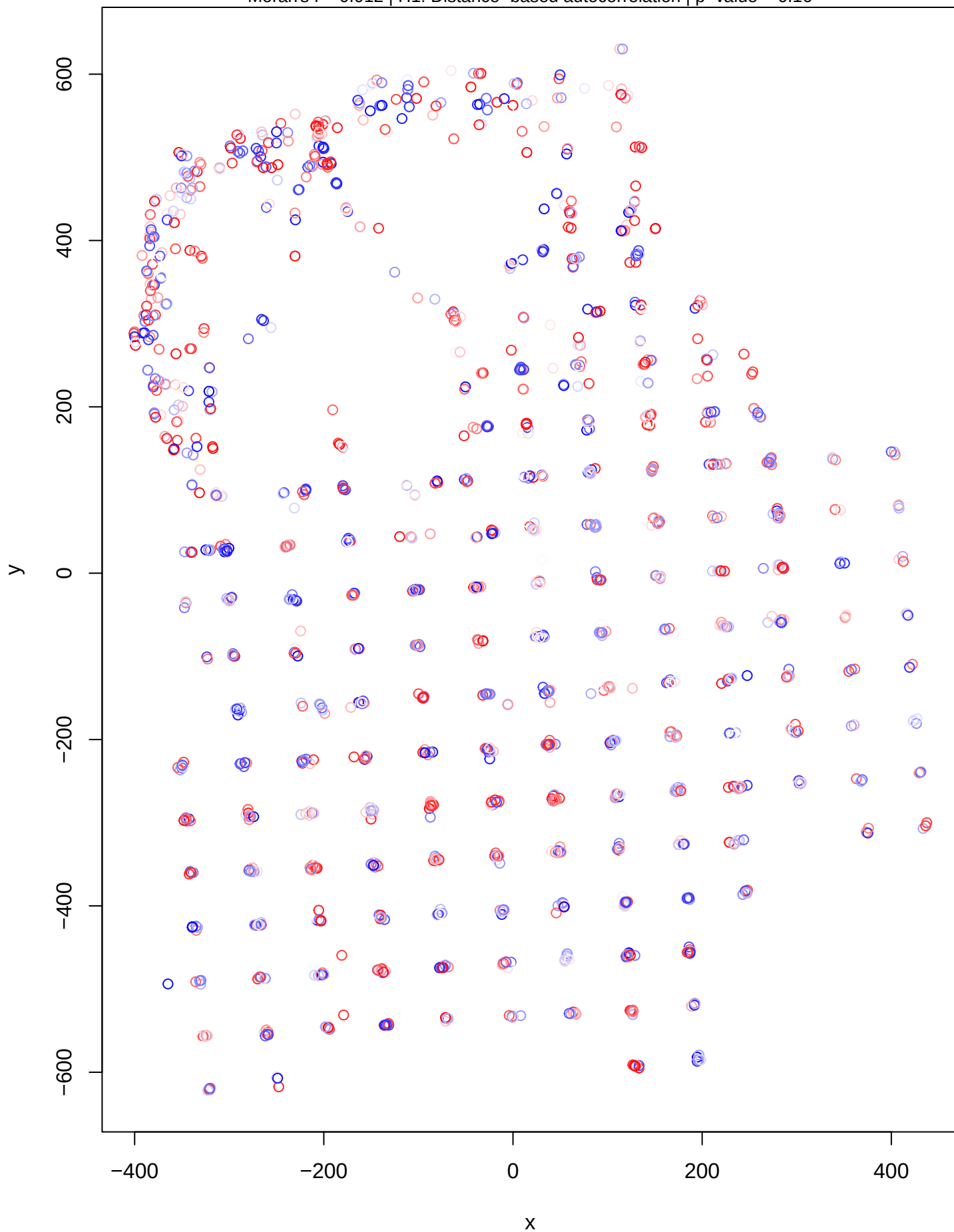
QQ plot residuals



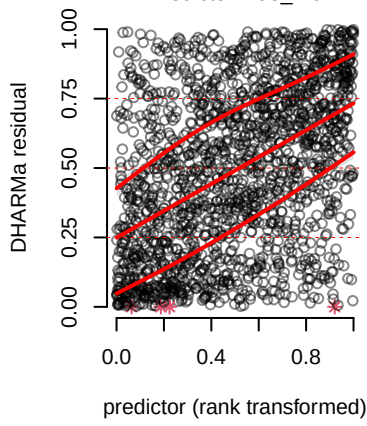
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



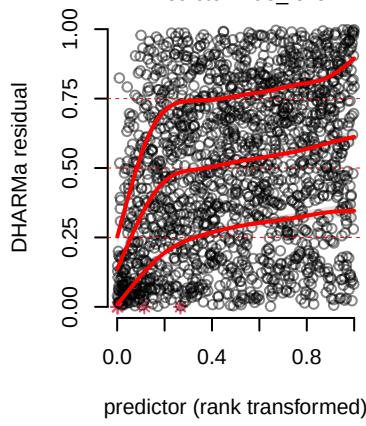
DHARMA Moran's I test for distance-based autocorrelation
Moran's I = 0.012 | H1: Distance-based autocorrelation | p-value = 0.16



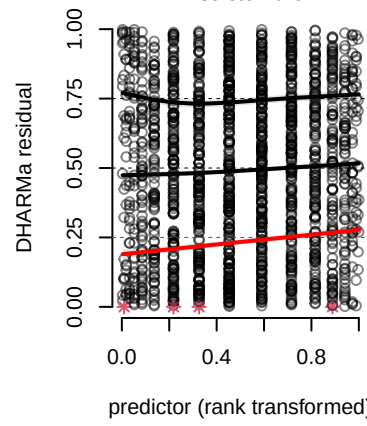
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



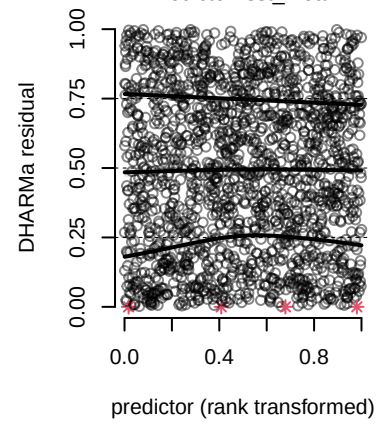
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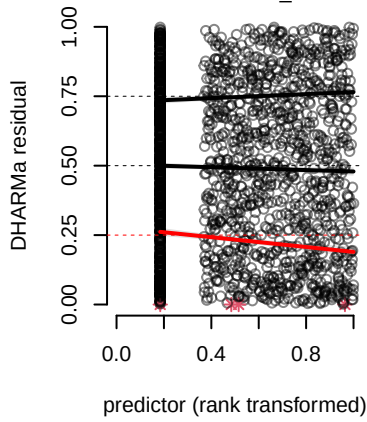
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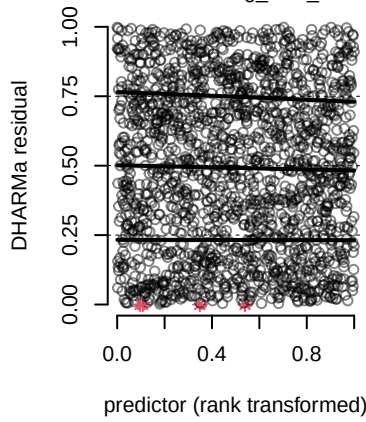
DHARMA residual Residual vs. predictor
No significant problems detected



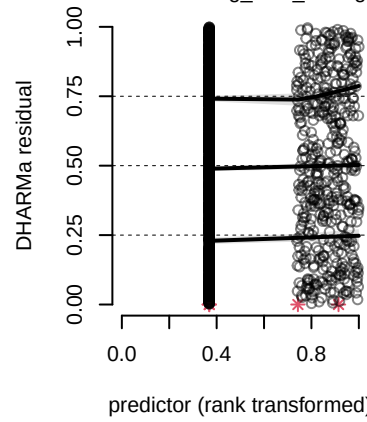
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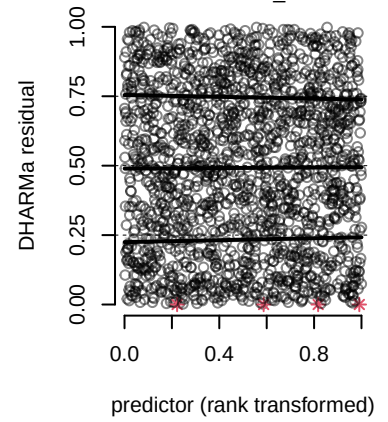
DHARMA residual Residual vs. predictor
No significant problems detected



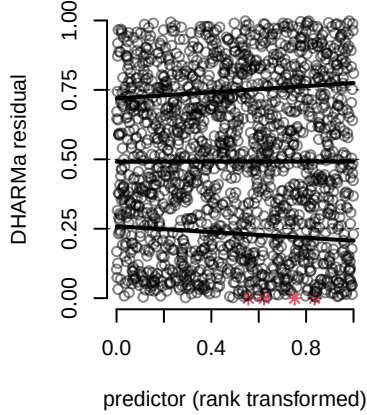
DHARMA residual Residual vs. predictor
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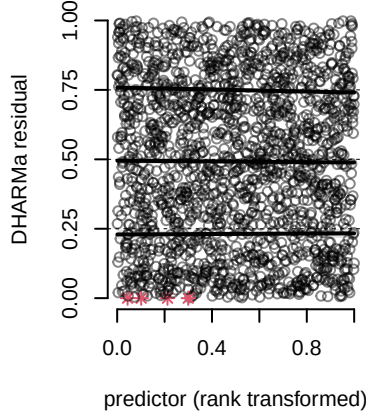
DHARMA residual Residual vs. predictor
No significant problems detected



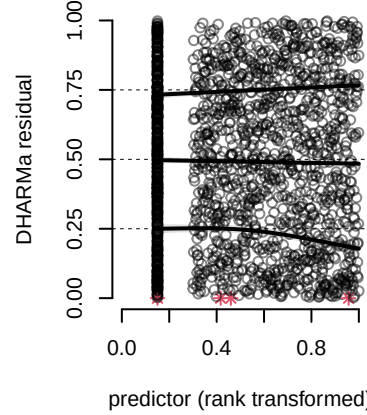
DHARMA residual Residual vs. predictor
No significant problems detected



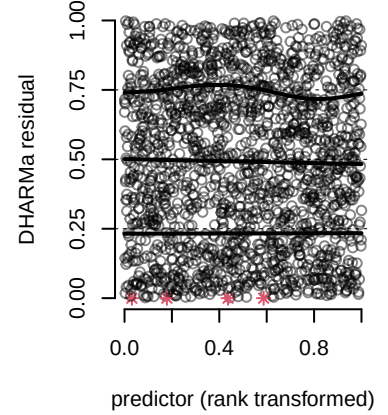
DHARMA residual Residual vs. predictor
No significant problems detected



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DHARMA residual Residual vs. predictor
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